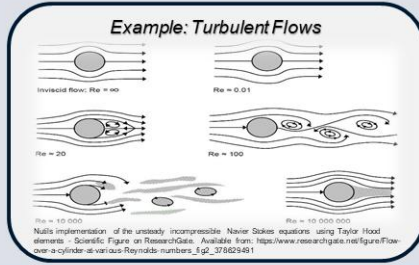


Turbulence Estimation by Laser Imaging

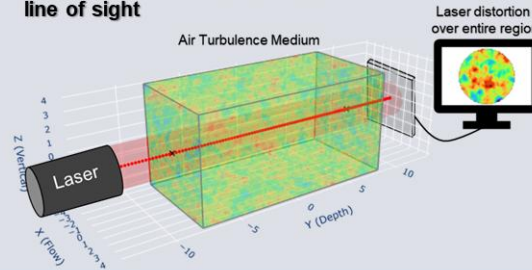
¹Karl Weisenburger, ¹Gregery Buzzard, ²Charles Bouman, ³Matthew Kemnetz

¹Mathematics, Purdue University, ²Electrical and Computer Engineering, and Biomedical Engineering, Purdue University ³Engineering Physics, Air Force Institute of Technology

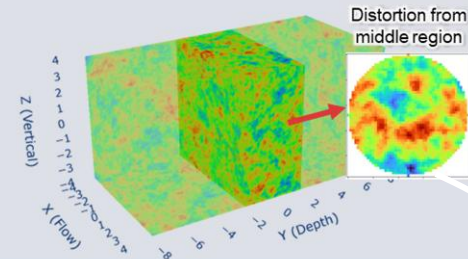
Big Goal: Estimate Turbulence in a Volume



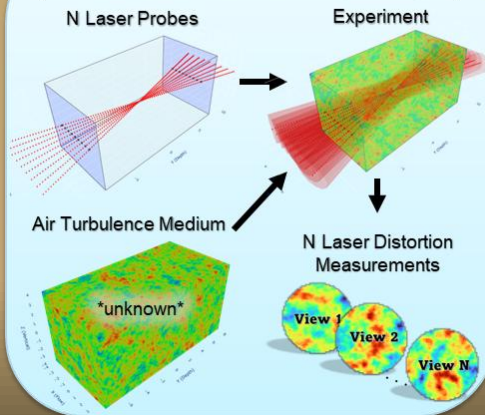
Problem: Laser imaging integrates distortions along line of sight



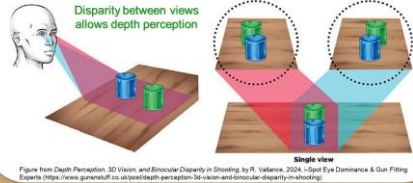
Subgoal: Estimate turbulence at single depth



Idea: Use Views from Multiple Angles



Binocular Vision: How do humans perceive depth?



Approach: Jointly optimize for fit to data and smooth variations

"Find a **smooth volume** that fits the **collected data**"

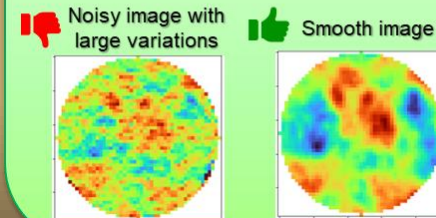
$$\arg \min \left\{ \left\| \begin{array}{c} \text{Simulated laser distortion from reconstruction} \\ \text{Measured laser distortion from ground truth} \end{array} \right\|^2 + GGMRF \left(\begin{array}{c} \text{Reconstruction} \end{array} \right) \right\}$$

Data fitting
 Prior Fitting (smoothing)

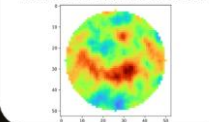
Prior Model:

Generalized Gaussian Markov Random Field (GGMRF)

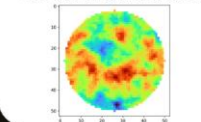
Penalizes Large Variations



Reconstructed Distortion



Ground Truth Distortion



Issues with real data:

- > Missing projection information
- > Noisy measurements
- > Limited views
- > Limited angular diversity
- > Narrow field of view

Math techniques used:

- > Maximum A Posteriori estimation
- > Nuisance parameter estimation
- > View selection optimization
- > Vectorized coordinate descent
- > High performance computing

Journal Paper

